GRlite

A Real-Time 2D Linearized-Gravity + Electromagnetism
Particle-in-Cell Sandbox

Technical Report v1

D. Marks
`github.com/profdc9/GRlite`

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Abstract

GRlite is an interactive sandbox that integrates *linearized* general relativity coupled to electromagnetism on a 2D grid, in real time, in a web browser. It is a particle-in-cell (PIC) field solver: massive and charged macroparticles source six gravito-electromagnetic potentials on a Yee lattice, the potentials are advanced by an explicit wave-equation leapfrog, and the particles are pushed by a relativistic force law gathered from those potentials. This report documents the simulator at the level of detail of a technical report, in three parts. **Part I** derives the physical model: linearized GR in the gravito-electromagnetic (GEM) form, its coupling to Maxwell electromagnetism, the relativistic single-particle dynamics (including post-Newtonian corrections, spin, and proper time), and the background/perturbation split. Each result is derived in 3D and then specialized to the 2+1 dimensions the simulator actually runs in, so the correspondence between the full physics and the reduced model is explicit. **Part II** (separate) builds the PIC scheme from the 2D equations: Yee sublattices and the leapfrog field update; the CIC/TSC/bump shape functions; charge-conserving Esirkepov current deposition; the Lewis–Birdsall momentum-conserving force gather; the relativistic Boris pusher; initial-field generation; the absorbing boundary; and the visualization mathematics. **Part III** (separate) documents the C library that implements the scheme and maps every public function to the step of the PIC algorithm it realizes.

Contents

Conventions, units, and notation

Signature and indices. The metric signature is $(-,+,+,+)$. Greek indices $\mu,\nu,\ldots$ run over $0,1,2,3$ (spacetime); Latin indices i,j,k over the spatial dimensions. Bold symbols ($\mathbf{x}$, $\mathbf{v}$, $\mathbf{A}_g$) are spatial vectors — 3-vectors in the derivations of this Part, 2-vectors after the 2D reduction.

Effective constants. The simulator does not use SI values for the fundamental constants; it exposes *tunable* effective constants so a student can dial gravity, electrostatics, and the wave speed independently and explore limits ($c \to \infty$ recovers the Newtonian limit; $G_{\text{eff}} \to 0$ switches off gravity; etc.). Throughout this report the symbols G, c, and the Coulomb constant $k_e = 1/(4\pi\varepsilon_0)$

denote those effective quantities G_{eff}, c_{eff}, k_e. We write G and c for readability and flag the substitution where it matters numerically (e.g. the Schwarzschild precession rate $\propto G_{\text{eff}} M / c_{\text{eff}}^2$).

The two object classes. GRlite distinguishes two kinds of gravitating/charged object: *macroparticles*, which move, deposit fields on the grid, and feel forces; and *background bodies*, which are fixed analytic sources (a precomputed field that never moves and feels no back-reaction). Part I treats their common physics; the split itself is the subject of §5.

The six potentials. The dynamical state of the field is six scalar potentials,

$$\underbrace{\Phi_g,\ A_{g,x},\ A_{g,y}}_{\text{gravity (GEM)}}, \qquad \underbrace{\phi,\ A_x,\ A_y}_{\text{electromagnetism}}, \tag{1}$$

each advanced by a sourced wave equation. Derived fields ($\mathbf{g}$, $\mathbf{B}_g$, $\mathbf{E}$, $\mathbf{B}$) are never stored; they are reconstructed on demand from finite differences of these potentials. This "potentials are the state" choice drives much of both the physics formulation (Part I) and the numerics (Part II).

Part I
Physical Foundations

1 Linearized General Relativity and Gravito-electromagnetism

1.1 Goal and regime

A full numerical-relativity evolution (e.g. the ADM 3+1 split with a Cauchy boundary) is far too expensive for an interactive desktop/browser tool. GRlite instead works in *linearized* GR, where the metric is a small perturbation of flat spacetime,

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \qquad |h_{\mu\nu}| \ll 1. \tag{2}$$

In the Lorenz gauge $\partial_\mu \bar{h}^{\mu\nu} = 0$, the trace-reversed perturbation $\bar{h}^{\mu\nu} \equiv h^{\mu\nu} - \frac{1}{2}\eta^{\mu\nu} h$ obeys a sourced wave equation,

$$\Box\, \bar{h}^{\mu\nu} = -\frac{16\pi G}{c^4}\, T^{\mu\nu}_{\text{total}}, \qquad \Box \equiv -\frac{1}{c^2}\partial_t^2 + \nabla^2. \tag{3}$$

This is structurally identical to the wave equation for the electromagnetic 4-potential, which is exactly why FDTD methods from computational electromagnetics carry over directly — and why the whole simulator can treat gravity and electromagnetism with one field engine.

1.2 The GEM metric

Choosing the gravito-electromagnetic (GEM) potentials Φ_g (gravitoelectric scalar) and $\mathbf{A}_g$ (gravitomagnetic vector) to carry the relevant components of $\bar{h}^{\mu\nu}$, the linearized metric is

$$h_{00} = -\frac{2\Phi_g}{c^2}, \qquad h_{0i} = \frac{2A_{g,i}}{c}, \qquad h_{ij} = -\frac{2\Phi_g}{c^2}\,\delta_{ij}, \tag{4}$$

with line element

$$\mathrm{d}s^2 = -\Big(1 + \tfrac{2\Phi_g}{c^2}\Big)c^2\,\mathrm{d}t^2 + \frac{4}{c}\,\mathbf{A}_g \cdot \mathrm{d}\mathbf{x}\,\mathrm{d}t + \Big(1 - \tfrac{2\Phi_g}{c^2}\Big)(\mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2). \tag{5}$$

The trace-reversed components are $\bar{h}_{00} = -4\Phi_g/c^2$, $\bar{h}_{0i} = 4A_{g,i}/c$, $\bar{h}_{ij} = 0$, so Eq. (3) decouples cleanly into $\Box\Phi_g \propto \rho$ and $\Box\mathbf{A}_g \propto \mathbf{J}$ with no spatial-stress source at this order.

What each metric component does. The three distinct pieces of Eq. (4) are responsible for disjoint classes of effect, and the simulator can switch them independently:

- $h_{00} = -2\Phi_g/c^2$ (time–time): the Newtonian potential. Gives gravitational attraction $-\nabla\Phi_g$, gravitational time dilation ($+\Phi_g/c^2$ in $\mathrm{d}\tau/\mathrm{d}t$), the Shapiro delay, and *half* of light bending.
- $h_{ij} = -2\Phi_g/c^2\,\delta_{ij}$ (spatial): no Newtonian analogue and invisible to slow particles, but supplies the *second half* of light bending (recovering the GR $4GM/c^2b$), the post-Newtonian force corrections (§4.2), and hence perihelion precession.
- $h_{0i} = 2A_{g,i}/c$ (time–space): nonzero only for moving/rotating mass currents. Produces the gravitomagnetic force, Lense–Thirring gyroscope precession $\mathbf{\Omega}_{\rm LT} = \nabla\times\mathbf{A}_g$, the gravitomagnetic clock effect (§4.4), and frame dragging.

1.3 GEM field equations and the Lorentz force

In the slow-motion weak-field limit the linearized equations take Maxwell form for a gravitoelectric field $\mathbf{g}$ and gravitomagnetic field $\mathbf{B}_g$:

$$\nabla\cdot\mathbf{g} = -4\pi G\rho, \qquad \nabla\times\mathbf{g} = -\partial_t\mathbf{B}_g, \tag{6}$$

$$\nabla\cdot\mathbf{B}_g = 0, \qquad \nabla\times\mathbf{B}_g = -\frac{4\pi G}{c^2}\mathbf{J} + \frac{1}{c^2}\partial_t\mathbf{g}, \tag{7}$$

with $\mathbf{g} = -\nabla\Phi_g - \partial_t\mathbf{A}_g$ and $\mathbf{B}_g = \nabla\times\mathbf{A}_g$. A test particle of rest mass m and charge q feels the combined GEM+EM Lorentz force

$$\mathbf{F} = m\big(\mathbf{g} + 4\,\mathbf{v}\times\mathbf{B}_g\big) + q\big(\mathbf{E} + \mathbf{v}\times\mathbf{B}\big). \tag{8}$$

The factor of 4 on the gravitomagnetic term is the fingerprint of the spin-2 graviton (versus spin-1 photon) and is responsible for the Lense–Thirring rate; getting it right also fixes the metric coefficient $h_{0i} = 2A_{g,i}/c$ and the proper-time cross term (§4.4).

1.4 Potential-form force: what the code actually evaluates

Because the simulator stores *potentials*, not fields, it is essential to rewrite Eq. (8) so that no grid-wide curl is required. Using $\mathbf{v}\times(\nabla\times\mathbf{A}) = \nabla(\mathbf{v}\cdot\mathbf{A}) - (\mathbf{v}\cdot\nabla)\mathbf{A}$,

$$\mathbf{F}_{\rm GEM} = m\big(-\nabla\Phi_g - \partial_t\mathbf{A}_g + 4\nabla(\mathbf{v}\cdot\mathbf{A}_g) - 4(\mathbf{v}\cdot\nabla)\mathbf{A}_g\big), \tag{9}$$

$$\mathbf{F}_{\rm EM} = q\big(-\nabla\phi - \partial_t\mathbf{A} + \nabla(\mathbf{v}\cdot\mathbf{A}) - (\mathbf{v}\cdot\nabla)\mathbf{A}\big). \tag{10}$$

Every term is a spatial or temporal derivative of a stored potential evaluated at the particle. The force is gauge-invariant: under $\phi\to\phi-\partial_t\Lambda$, $\mathbf{A}\to\mathbf{A}+\nabla\Lambda$ the per-term gauge dependence cancels. These two expressions (with the relativistic corrections of §4.2 added to the gravity piece) are the force law the pusher of Part II implements.

2D specialization. In 2+1 dimensions $\mathbf{g}$ and $\mathbf{E}$ are 2-vectors; $\mathbf{B}_g$ and $\mathbf{B}$ collapse to the single scalar curl components $B_{g,z} = \partial_x A_{g,y} - \partial_y A_{g,x}$ and $B_z = \partial_x A_y - \partial_y A_x$. Cross products become scalar multiplies, $(\mathbf{v}\times\mathbf{B}_g) = (v_y B_{g,z},\ -v_x B_{g,z})$, so the field-form force (8) reads $F_x = m(g_x + 4v_y B_{g,z}) + q(E_x + v_y B_z)$, $F_y = m(g_y - 4v_x B_{g,z}) + q(E_y - v_x B_z)$. The potential-form expressions (9)–(10) are unchanged in form with $\nabla = (\partial_x, \partial_y)$.

1.5 Particle properties: mass and charge are independent

Each macroparticle carries a rest mass m and an electric charge q that couple to disjoint fields:

- rest mass m sources gravity, $\rho_{\text{matter}} = \sum_i \gamma m_i \, \delta(\mathbf{x} - \mathbf{x}_i)$ and $\mathbf{J}_{\text{matter}} = \sum_i \gamma m_i \mathbf{v}_i \, \delta(\mathbf{x} - \mathbf{x}_i)$, and sets the inertia via $\mathbf{p} = \gamma m \mathbf{v}$;

- charge q sources EM, $\rho_q = \sum_i q_i \, \delta(\mathbf{x} - \mathbf{x}_i)$ and $\mathbf{J}_q = \sum_i q_i \mathbf{v}_i \, \delta(\mathbf{x} - \mathbf{x}_i)$, and sets the response to $q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$.

A particle with both $m \neq 0$ and $q \neq 0$ contributes to all four potential families and feels all four force families.

2 Electromagnetism in the Linearized Background

2.1 Potentials and wave equations

Maxwell's equations in the Lorenz gauge give the same wave-equation structure as gravity,

$$\Box \phi = -\rho_q / \varepsilon_0, \qquad \Box \mathbf{A} = -\mu_0 \mathbf{J}_q, \tag{11}$$

with $\mathbf{E} = -\nabla \phi - \partial_t \mathbf{A}$ and $\mathbf{B} = \nabla \times \mathbf{A}$. In the GRlite engine these are advanced by the *same* leapfrog as the GEM potentials; the only difference is the source and the local wave speed below.

2.2 Shapiro delay and gravitational lensing via c_{local}

A photon follows a null geodesic, $\mathrm{d}s^2 = 0$. Substituting the line element (5) (with both the time and the spatial coefficient) and solving for the coordinate light speed gives

$$\frac{\mathrm{d}x}{\mathrm{d}t} \approx c\Big(1 + \frac{2\Phi_g}{c^2}\Big) \equiv c_{\text{local}}. \tag{12}$$

The factor of 2 comes from both metric coefficients contributing to the null condition; it is the same factor that doubles light bending over the Newtonian value. The Shapiro time delay is $\Delta t = -\frac{2}{c^3} \int_{\text{path}} \Phi_g \, \mathrm{d}\ell$. GRlite realizes both Shapiro delay and lensing by *stepping the EM wave equation with* $c_{\text{local}}(\mathbf{x})$ *in place of* c; wavefronts slow and bend in the potential well exactly as light should. Crucially Φ_g here is the *total* (background + perturbation) potential, so a moving deposited mass can lens an EM wave (a dynamic lens).

2.3 Electromagnetic stress-energy as a gravitational source

Electromagnetic field energy gravitates. Writing $\mathbf{P} \equiv \nabla \phi + \partial_t \mathbf{A} = -\mathbf{E}$ and $\mathbf{C} \equiv \nabla \times \mathbf{A} = \mathbf{B}$, the EM contributions to the GEM source (T^{00} energy density and T^{0i} Poynting flux, divided by c^2) are

$$\rho_{\text{EM}} = \frac{\varepsilon_0}{2c^2} \big(|\mathbf{P}^{\text{pert}}|^2 + c^2 |\mathbf{C}^{\text{pert}}|^2\big), \tag{13}$$

$$\mathbf{J}_{\text{EM}} = -\frac{1}{\mu_0 c^2} \, \mathbf{P}^{\text{pert}} \times \mathbf{C}^{\text{pert}}. \tag{14}$$

Only the *perturbation* potentials enter, to avoid double-counting the background metric. The Maxwell spatial stress T^{ij} is dropped (it would source h^{ij}, which the GEM reduction does not evolve); this is the principal approximation of the "effective-source" method and is valid when EM fields are not so strong or directional that T^{ij} matters ($|\mathbf{S}|/c \ll u_{\text{EM}}$).

2D specialization. The curl becomes the scalar $F_{12}^{\text{pert}} = \partial_x A_y^{\text{pert}} - \partial_y A_x^{\text{pert}}$, and the cross product in (14) becomes a 90° rotation of the 2-vector $\mathbf{P}^{\text{pert}}$ scaled by F_{12}^{pert}:

$$\rho_{\text{EM}} = \frac{\varepsilon_0}{2c^2}\left(|\mathbf{P}^{\text{pert}}|^2 + c^2(F_{12}^{\text{pert}})^2\right), \quad \mathbf{J}_{\text{EM}} = -\frac{F_{12}^{\text{pert}}}{\mu_0 c^2}\begin{pmatrix} P_y^{\text{pert}} \\ -P_x^{\text{pert}} \end{pmatrix}. \tag{15}$$

3 The Unified Six-Field System

Collecting §§1–2, the dynamical field state and its sources are summarized in Table 1. All six potentials obey the *same* wave operator; gravity and EM differ only in coupling sign/strength and (for EM) the use of c_{local}. This uniformity is what lets a single leapfrog kernel (Part II) advance the entire system.

Potential	**Wave equation**	**Source**	**Derived field**
Φ_g	$\Box\Phi_g = -4\pi G\rho_{\text{eff}}$	ρ_{eff}	$\mathbf{g} = -\nabla\Phi_g - \partial_t\mathbf{A}_g$
$A_{g,x}, A_{g,y}$	$\Box\mathbf{A}_g = -\frac{4\pi G}{c^2}\mathbf{J}_{\text{eff}}$	$\mathbf{J}_{\text{eff}}$	$B_{g,z} = (\nabla\times\mathbf{A}_g)_z$
ϕ	$\Box_{c_{\text{local}}}\phi = -\rho_q/\varepsilon_0$	ρ_q	$\mathbf{E} = -\nabla\phi - \partial_t\mathbf{A}$
A_x, A_y	$\Box_{c_{\text{local}}}\mathbf{A} = -\mu_0\mathbf{J}_q$	$\mathbf{J}_q$	$B_z = (\nabla\times\mathbf{A})_z$

Table 1: The six-potential system. $\rho_{\text{eff}} = \rho_{\text{matter}} + \rho_{\text{EM}}$ and $\mathbf{J}_{\text{eff}} = \mathbf{J}_{\text{matter}} + \mathbf{J}_{\text{EM}}$ fold the EM stress-energy (13)–(14) back into the gravitational source. In 2D the two vector potentials per family are genuine 2-vectors and the curls are scalars.

4 Relativistic Particle Dynamics

4.1 Relativistic momentum update

Particles are integrated through the relativistic 3-momentum $\mathbf{p} = \gamma m\mathbf{v}$, $\gamma = (1 - v^2/c^2)^{-1/2}$:

$$\mathbf{p} \leftarrow \mathbf{p} + \mathbf{F}_{\text{total}}\,dt, \qquad \mathbf{v} = \frac{\mathbf{p}}{\sqrt{m^2 + |\mathbf{p}|^2/c^2}}, \qquad \mathbf{x} \leftarrow \mathbf{x} + \mathbf{v}\,dt. \tag{16}$$

Pushing $\mathbf{p}$ (not $\mathbf{v}$) automatically captures the anisotropy of relativistic inertia (longitudinal mass $\gamma^3 m$ vs. transverse γm). The EM force (10) is already exactly relativistic; the gravitational force needs the corrections below.

4.2 Post-Newtonian (EIH) corrections

GEM as written is a slow-motion expansion: at relativistic speed the spatial metric h^{ij} also deflects particles (this is the missing "second half" of light bending). The correct $\mathcal{O}((v/c)^2)$ equation of motion is the test-particle limit of the Einstein–Infeld–Hoffmann (EIH) 1PN equation [1, 2]. Writing $\mathbf{g} = -\nabla\Phi_g - \partial_t\mathbf{A}_g$ and organizing by order in v/c,

$$\mathbf{a} = \underbrace{\mathbf{g}}_{\mathcal{O}(1)} + \underbrace{4\,\mathbf{v}\times\mathbf{B}_g}_{\mathcal{O}(v/c)\ \text{(Lense–Thirring)}} + \underbrace{\frac{v^2}{c^2}\mathbf{g} + \frac{4\Phi_g}{c^2}\mathbf{g} - \frac{4(\mathbf{v}\cdot\mathbf{g})\mathbf{v}}{c^2}}_{\mathcal{O}(v^2/c^2)\ \text{(EIH 1PN)}} + \mathcal{O}(v^3/c^3). \tag{17}$$

The three 1PN terms are physically distinct: $(v^2/c^2)\mathbf{g}$ strengthens deflection of fast particles (and, with the $c_{\rm local}$ spatial-metric piece, recovers the factor-of-2 photon deflection as $v \to c$); $(4\Phi_g/c^2)\mathbf{g}$ is a radial "Shapiro" enhancement contributing to binding energy and precession; and $-4(\mathbf{v}\cdot\mathbf{g})\mathbf{v}/c^2$ is a velocity-coupling term that does no net work but breaks Keplerian closure. Integrated over a bound orbit they reproduce the Schwarzschild perihelion advance

$$\Delta\varphi_{\rm prec} = \frac{6\pi\, G_{\rm eff} M}{c_{\rm eff}^2\, a(1-e^2)}. \tag{18}$$

The simulator exposes these as force *tiers* (Newtonian; GEM; EIH 1PN), selectable per scenario; Table 2 lists them. In implementation the gravitomagnetic cross product is always replaced by its potential form (§1.4), so $\mathbf{B}_g$ is never formed as a grid array; the scalar $\mathbf{g}$ used in the 1PN terms is evaluated on the fly from the potentials at the particle.

Tier	**Gravitational acceleration**	**Range**
Newtonian	$\mathbf{g}$	$v/c < 0.01$
GEM	$\mathbf{g} + 4\,\mathbf{v}\times\mathbf{B}_g$ (potential form)	$v/c < 0.1$
EIH 1PN	GEM + $\frac{v^2}{c^2}\mathbf{g} + \frac{4\Phi_g}{c^2}\mathbf{g} - \frac{4(\mathbf{v}\cdot\mathbf{g})\mathbf{v}}{c^2}$	$v/c \lesssim 0.5$

Table 2: Gravitational force tiers. EM is exactly relativistic in all tiers; the momentum update (16) is always relativistic.

4.3 Intrinsic spin

A particle with intrinsic angular momentum obeys the Mathisson–Papapetrou–Dixon equations; in the GEM/EM approximation these reduce to a 3-vector spin $\mathbf{S}$ integrated alongside the momentum. The spin obeys the Tulczyjew–Dixon condition $S^{\mu\nu}p_\nu = 0$ (note: p^μ, the momentum, not mU^μ), whose 3+1 split fixes $S^0 = \mathbf{S}\cdot\mathbf{p}/(\gamma mc)$ and is maintained each step by the covariant projection

$$\mathbf{S} \leftarrow \mathbf{S} - \frac{\mathbf{S}\cdot\mathbf{p}}{|\mathbf{p}|^2 + m^2c^2}\,\mathbf{p}. \tag{19}$$

Spin precesses under gravitomagnetic, EM, and Thomas terms,

$$\frac{\mathrm{d}\mathbf{S}}{\mathrm{d}t} = \mathbf{S}\times\Big(\underbrace{\nabla\times\mathbf{A}_g}_{\text{grav}} + \underbrace{\tfrac{g_s q}{2m}\nabla\times\mathbf{A}}_{\text{EM}} + \underbrace{-\tfrac{\gamma^2}{\gamma+1}\tfrac{\mathbf{v}\times\mathbf{a}}{c^2}}_{\text{Thomas}}\Big), \tag{20}$$

and, being a moving dipole, sources magnetization currents (magnetic moment $\boldsymbol{\mu} = g_s(q/2m)\mathbf{S}$, gravitomagnetic moment $\boldsymbol{\sigma} = 2\mathbf{S}$). A spin gradient also exerts a Stern–Gerlach force $\mathbf{F}_{\rm spin} = \nabla(\mathbf{S}\cdot\mathbf{B}_g) + \nabla(\boldsymbol{\mu}\cdot\mathbf{B})$.

2D specialization. With motion in the plane and spin along $\hat{z}$, $\mathbf{S}$ is a single scalar s (an angle $\phi_{\rm spin}$), and $\nabla\times\mathbf{A}_g$, $\nabla\times\mathbf{A}$ are the scalars $B_{g,z}$, B_z. Precession reduces to $\dot{\phi}_{\rm spin} = B_{g,z} + \frac{g_s q}{2m}B_z + \Omega_{{\rm Thomas},z}$ with $\Omega_{{\rm Thomas},z} = -\frac{\gamma^2}{\gamma+1}(v_x a_y - v_y a_x)/c^2$, the spin-gradient force becomes $s\,\nabla B_{g,z} + \mu\,\nabla B_z$, and the Tulczyjew–Dixon constraint (19) is satisfied automatically (no projection step needed).

4.4 Proper time

Each particle integrates the proper time along its worldline from the metric (5), $\mathrm{d}\tau^2 = -\mathrm{d}s^2/c^2$:

$$\frac{\mathrm{d}\tau}{\mathrm{d}t} = \sqrt{\Big(1+\frac{2\Phi_g}{c^2}\Big) - \Big(1-\frac{2\Phi_g}{c^2}\Big)\frac{v^2}{c^2} - \frac{8\,\mathbf{A}_g\cdot\mathbf{v}}{c^2}} \;\approx\; 1+\frac{\Phi_g}{c^2}-\frac{v^2}{2c^2}-\frac{4\,\mathbf{A}_g\cdot\mathbf{v}}{c^2}. \tag{21}$$

The three first-order terms are the gravitational redshift (Φ_g/c^2, slows clocks in a well), the kinematic dilation ($-v^2/2c^2$), and the *gravitomagnetic clock effect* ($-4\mathbf{A}_g\cdot\mathbf{v}/c^2$). The cross-term coefficient is fixed to 8 inside the root (equivalently 4 at first order) by consistency with the factor-of-4 force law / $h_{0i} = 4A_{g,i}/c$ trace-reversed component; this gives prograde clocks running *slower* than retrograde at equal radius, matching the Cohen–Mashhoon Kerr result [3, 4]. Around a closed orbit the prograde−retrograde difference is the gravitomagnetic flux, $\Delta\tau = -\frac{8}{c^2}\oint\mathbf{A}_g\cdot\mathrm{d}\mathbf{x} = -\frac{8}{c^2}\int\mathbf{B}_g\cdot\mathrm{d}\mathbf{A}$. The same interpolated scalar $\mathbf{v}\cdot\mathbf{A}_g$ already computed for the force is reused here, so a proper-time clock costs one square root per particle per step.

5 Background Spacetime and the Perturbation Split

5.1 Decomposition and modes

When fixed sources dominate the field, it is wasteful to evolve them on the grid. GRlite splits every potential into a precomputed analytic *background* and a time-stepped *perturbation*,

$$\Phi_g(\mathbf{x},t) = \Phi_g^{\mathrm{bg}}(\mathbf{x}) + \Phi_g^{\mathrm{pert}}(\mathbf{x},t), \tag{22}$$

and likewise for $\mathbf{A}_g, \phi, \mathbf{A}$. The force on a particle uses the sum of both contributions; only the perturbation is advanced by the wave solver, and it is sourced only by the moving macroparticles. Three modes follow: pure background (test particles, zero field cost), background + perturbation (a few interacting particles in a dominant field), and fully dynamic (no background; all fields evolved, e.g. an equal-mass binary).

5.2 The compact body: a no-hair (M,Q,J) source

The canonical background is a single fixed *compact body* carrying the three "no-hair" charges of a stationary black hole — mass M, electric charge Q, angular momentum J_z — superposed in one object. Its analytic potentials are the softened point sources

$$\Phi_g^{\mathrm{bg}} = -\frac{GM}{\sqrt{r^2+\varepsilon^2}}, \qquad \phi^{\mathrm{bg}} = +\frac{k_eQ}{\sqrt{r^2+\varepsilon^2}}, \tag{23}$$

$$\mathbf{A}_g^{\mathrm{bg}} = \frac{GJ_z}{2c^2}\,\frac{\hat{z}\times\mathbf{r}}{(r^2+\varepsilon^2)^{3/2}}, \qquad \mathbf{A}^{\mathrm{bg}} = \frac{k_e\,\mu_z}{c^2}\,\frac{\hat{z}\times\mathbf{r}}{(r^2+\varepsilon^2)^{3/2}}, \tag{24}$$

where $r = |\mathbf{x}-\mathbf{x}_0|$ and ε is a softening length (a few cells) that removes the singular core. The special cases reproduce the weak-field analogues of the classical metrics: $(M,0,0)$ Schwarzschild, $(M,Q,0)$ Reissner–Nordström, $(M,0,J)$ Kerr, (M,Q,J) Kerr–Newman.

The magnetic dipole is fixed by the no-hair relation. A spinning charge has an EM magnetic dipole moment set by the gyromagnetic relation

$$\mu_z = g_s\,\frac{Q}{2M}\,J_z = \frac{g_s\,Q\,J_z\,G_{\mathrm{eff}}}{2\,GM}, \tag{25}$$

with $g_s = 2$ reproducing the Kerr–Newman value $\mu = QJ/M$. Equation (25) is undefined for $M = 0$, so a massless charged rotor carries *no* EM dipole field — the correct degenerate limit, since such an object is not a Kerr–Newman body. (The gravitomagnetic dipole $\mathbf{A}_g^{\mathrm{bg}}$, by contrast, is sourced by J_z alone and is mass-independent.)

Superposition. Multiple compact bodies superpose linearly: the background potential is the sum of each body's analytic field. This is exact in the linearized theory (the field equations are linear), and is the basis for multi-body scenes — the two-fixed-center problem, multi-lens gravitational lensing, and mixed $M/Q/J$ fields. The simulator supports up to a fixed number of such bodies (Part III).

5.3 The 2D Green's function choice

In strict 2D the static Green's function of ∇^2 is logarithmic, so a 2D point mass gives $\Phi_g \sim \ln r$ and $|\mathbf{g}| \sim 1/r$ (flat rotation curves) rather than the 3D $\Phi_g \sim 1/r$, $|\mathbf{g}| \sim 1/r^2$:

$$\text{3D: } \Phi_g \sim 1/r,\ |\mathbf{g}| \sim 1/r^2; \qquad \text{2D: } \Phi_g \sim \ln r,\ |\mathbf{g}| \sim 1/r. \tag{26}$$

GRlite makes a deliberate pedagogical choice here: the *deposited* (perturbation) field is the honest 2D solver output (a $\ln r$ field, since it solves the 2D Poisson/wave equation), while the analytic *background* body prescribes the 3D-flavored softened $1/r$ form of §5.2, so that test-particle orbits look Keplerian and the precession test (18) is meaningful. The FDTD perturbation solver is identical in either case; only the analytic background differs. This dual nature (true-2D deposited fields vs. 3D-flavored fixed sources) is intrinsic to the model and is flagged again where it affects the numerics (Part II) and the visualizations.

6 What Survives in 2D: Summary

Table 3 consolidates the 2D reduction used throughout Part I and assumed by Parts II–III. The headline is that essentially all of the physically interesting GR/EM effects survive the reduction; what is lost is the transverse tensor (gravitational-wave $+/\times$) polarizations, which require two transverse spatial directions absent in 2+1.

End of Part I. Part II builds the particle-in-cell numerical scheme directly on the 2D equations collected in Table 1 and §6; Part III documents the C library that implements it.

Part II
The Particle-in-Cell Scheme

7 Overview: the per-step pipeline

GRlite is a particle-in-cell (PIC) code: the continuous sources of Part I are represented by a finite set of macroparticles, deposited onto grid arrays each step; the six potentials are advanced by an explicit wave-equation leapfrog; and the particles are pushed by the force gathered back from those potentials. The dynamical state is

Quantity / effect	2D?	Reduced form / note
$\mathbf{g}, \mathbf{E}$	yes	2-vectors (x, y)
$\mathbf{B}_g, \mathbf{B}$	yes	scalars $B_{g,z}, B_z$ (the curl)
$\mathbf{v} \times \mathbf{B}_g$	yes	$(v_y B_{g,z}, -v_x B_{g,z})$
spin $\mathbf{S}$	yes	scalar angle ϕ_{spin}; TD constraint automatic
Newtonian gravity	yes	$\ln r$ deposited; $1/r$ available as background
frame dragging	yes	$B_{g,z}$ from spin/moving mass
perihelion precession	yes	$\mathcal{O}(v^2/c^2)$ EIH terms
Shapiro delay / lensing	yes	via c_{local}
proper-time dilation	yes	per-particle clock, Eq. (21)
gravitomagnetic clock effect	yes	$-8\mathbf{A}_g \cdot \mathbf{v}/c^2$ term
EM–gravity coupling	yes	$\rho_{\text{EM}}, \mathbf{J}_{\text{EM}}$
GW scalar mode	yes	propagates at c
GW tensor $(+, \times)$ modes	**no**	require h^{ij}; absent in 2D GEM

Table 3: The 2+1 reduction at a glance.

- six **potential** arrays $\{\Phi_g, A_{g,x}, A_{g,y}, \phi, A_x, A_y\}$, each carrying three time levels (prev $= {}^{n-1}$, curr $= {}^{n}$, next $= {}^{n+1}$) on its Yee sublattice (§8);
- six **source** arrays $\{\rho_{\text{matter}}, J_{m,x}, J_{m,y}, \rho_q, J_{q,x}, J_{q,y}\}$ rebuilt each step from the particles;
- a **particle** array, each entry storing $(x, y, p_x, p_y, m, q, \tau, \phi_{\text{spin}}, g_s, \dots)$;
- optional fixed **background** fields (§15).

One time step (the routine `gr_sim_step`, Part III) executes:

Step 1. Deposit sources. Clear the source arrays; for each particle deposit ρ^n at $\mathbf{x}^n$ with the chosen shape (§9), the current $\mathbf{J}^{n-1/2}$ over the segment $\mathbf{x}^{n-1} \to \mathbf{x}^n$ by Esirkepov (§10), and any spin-dipole curl current. Velocity is $\mathbf{v} = \mathbf{p}^{n-1/2}/(\gamma m)$. Optionally binomial-smooth $\rho, \mathbf{J}$ (§10).

Step 2. Add EM stress-energy to the gravitational sources, $\rho_{\text{matter}} \mathrel{+}= \rho_{\text{EM}}$, $\mathbf{J}_{\text{matter}} \mathrel{+}= \mathbf{J}_{\text{EM}}$ (Eqs. (13)–(14)).

Step 3. Refresh c_{local} from the current total Φ_g if dynamic Shapiro is on (§2).

Step 4. Advance the fields: one leapfrog sweep of all six potentials (§8).

Step 5. Absorb: apply the centered-implicit boundary friction to each `next` array (§14).

Step 6. Rotate the three time-level pointers of each field.

Step 7. Gauge fixes: optional zero-mean scalar subtraction and parabolic Lorenz cleaning (§14).

Step 8. Push particles: gather the force, Boris kick–drift, then update spin, proper time, and position (§12).

The leapfrog staggering puts particle momenta/velocities at half-integer times ($\mathbf{p}^{n-1/2}$) and positions/fields at integer times; this is what makes the deposition velocity, the force velocity, and the current-deposition segment mutually consistent.

8 The Yee Lattice and the Leapfrog Field Update

8.1 Sublattices

Each potential lives on its own staggered (Yee) sublattice so that the derived fields land where they are needed:

Potential	**Sublattice**	**Node position**
$\Phi_g,\ \phi$	corner	$(i,\ j)\,\Delta x$
$A_{g,x},\ A_x$	x-edge	$(i+\frac{1}{2},\ j)\,\Delta x$
$A_{g,y},\ A_y$	y-edge	$(i,\ j+\frac{1}{2})\,\Delta x$

Curls then live at cell centers $(i+\frac{1}{2}, j+\frac{1}{2})\Delta x$: e.g. $B_{g,z} = (A_{g,y}[i+1, j]-A_{g,y}[i, j])/\Delta x-(A_{g,x}[i, j+1]-A_{g,x}[i, j])/\Delta x$, which the force gather evaluates on the fly (§11). The five-point Laplacian at a field's own index sums neighbors on the *same* sublattice, so a single kernel form serves all six potentials; only the interpretation of (i, j) differs.

8.2 The discrete update

Writing $\Box\Psi = c^{-2}\partial_t^2\Psi - \nabla^2\Psi = \mathcal{S}$ for any of the six potentials, the explicit leapfrog is

$$\boxed{\Psi_k^{n+1} = 2\Psi_k^n - \Psi_k^{n-1} + (c\,\Delta t)^2\Big[(\nabla_h^2\Psi^n)_k + s_\Psi\,\mathcal{S}_k^n\Big], \quad (\nabla_h^2\Psi)_k = \frac{\Psi_{k\pm1} + \Psi_{k\pm W} - 4\Psi_k}{\Delta x^2}, \quad (27)}$$

where $k = jW + i$ is the flat array index and s_Ψ is the per-field source coupling: $s_{\Phi_g} = -4\pi G$, $s_{\mathbf{A}_g} = -4\pi G/c^2$ (gravity), and $s_\phi = 1/\varepsilon_0$, $s_{\mathbf{A}} = -\mu_0$ (EM), realizing the source rows of Table 1. The scheme is second-order in space and time and stable under the 2D Courant condition

$$\frac{c\,\Delta t}{\Delta x} \leq \frac{1}{\sqrt{2}}, \tag{28}$$

exposed as the dimensionless `cfl` parameter ($\Delta t = \text{cfl}\cdot\Delta x/c$; the library default is $\text{cfl} = 1/\sqrt{2}$).

8.3 Kernel variants

The same recurrence (27) appears in four forms, selected per run:

- **undamped** (interior default): Eq. (27) verbatim;
- **Shapiro** (EM fields only): c^2 on the Laplacian is replaced by the per-cell $c^2_{\text{local}}(\mathbf{x})$ array (§2), while the source term keeps the uniform c^2;
- **critical damping**: with per-cell $\gamma_k = 1-\sigma_k\Delta t$, $\Psi_k^{n+1} = 2\gamma_k\Psi_k^n-\gamma_k^2\Psi_k^{n-1}+(c\Delta t)^2[\nabla_h^2\Psi^n+s\mathcal{S}]$, whose double root at γ_k gives an exact per-step decay and reduces bit-for-bit to the undamped form where $\sigma_k = 0$;
- **periodic**: wrap-around indexing on all four edges (translation invariant).

For the non-periodic variants an outer-boundary post-pass sets the ring either to zero (Dirichlet) or to its inward neighbor (Neumann); §14 discusses the choice and the separate derivative-friction absorber.

9 Shape Functions and the Macroparticle

A macroparticle replaces the point $\delta(\mathbf{x}-\mathbf{x}_p)$ of Part I by a finite cloud $W(\mathbf{x}-\mathbf{x}_p)$. All deposition and interpolation use a separable shape $W(\mathbf{x}) = w(x)\,w(y)$ whose 1D factor is non-negative, symmetric, and a partition of unity, $\sum_i w(x-x_i) = 1$ for any x — the last property is what makes the deposited mass/charge independent of sub-cell position. GRlite offers three shapes, all on the corner sublattice for ρ and edge sublattices for $\mathbf{J}$:

CIC (W_2, bilinear, 2×2). With $\alpha = x_p/\Delta x - i_c$, $\beta = y_p/\Delta x - j_c$ the four weights are $(1-\alpha)(1-\beta)$, $\alpha(1-\beta)$, $(1-\alpha)\beta$, $\alpha\beta$, deposited as value $\times\, w/\Delta x^2$ (so $\sum_k \rho_k\, \Delta x^2 =$ value).

TSC (W_3, quadratic B-spline, 3×3). Anchored at the nearest node with $u = x_p/\Delta x - i_c \in [-\frac{1}{2}, \frac{1}{2})$,

$$w_{-1} = \tfrac{1}{2}\big(\tfrac{1}{2} - u\big)^2, \quad w_0 = \tfrac{3}{4} - u^2, \quad w_{+1} = \tfrac{1}{2}\big(\tfrac{1}{2} + u\big)^2, \tag{29}$$

a smoother 9-cell footprint with a continuous first derivative.

Bump (C^∞, radius-parameterized). A mollifier

$$b(d, R) = \exp\Big(\frac{-1}{1-(d/R)^2}\Big) \text{ for } |d| < R, \text{ else } 0, \qquad w_i = \frac{b(i-u, R)}{\sum_j b(j-u, R)}, \tag{30}$$

per-axis renormalized so the weights sum to one (hence charge-conserving). The window half-width is $\lceil R + \frac{1}{2} \rceil$ cells. Being C^∞ with compact support, its Fourier transform decays sub-exponentially, suppressing the grid-scale aliasing that drives PIC self-heating; the production setup uses the bump kernel at radius $R \approx 6$. The same w_i are reused for interpolation, and the analytic derivative $\mathrm{d}w_i/\mathrm{d}u$ (including the renormalization correction) is used for the gradient gather of §11.

The adjoint condition. Deposit and gather use the *same* shape; this Hockney–Eastwood adjoint pairing is what guarantees a particle exerts no net self-force in a uniform field, and underlies the momentum-conserving force gather (§11).

10 Charge-Conserving Current Deposition

10.1 Why exact continuity matters

The EM (and GEM) wave solver enforces no constraint that ties $\mathbf{J}$ to $\partial_t \rho$. If the deposited pair violates the *discrete* continuity equation

$$\frac{\rho_k^n - \rho_k^{n-1}}{\Delta t} + (\nabla_h \cdot \mathbf{J}^{\,n-1/2})_k = 0, \tag{31}$$

the excess charge sources a spurious longitudinal field that trails the particle as a self-force wake. GRlite therefore deposits $\mathbf{J}$ by the Esirkepov construction, which satisfies Eq. (31) *exactly*, to round-off, at every corner cell.

10.2 The Esirkepov split

Let S^{n-1}, S^n be the shape footprints at $\mathbf{x}^{n-1}, \mathbf{x}^n$, each a product $S_x S_y$. The 2D footprint change $\Delta S = S_x^n S_y^n - S_x^{n-1} S_y^{n-1}$ is split into orthogonal x- and y-flux parts

$$\Delta S_{ij}^x = (S_{x,i}^n - S_{x,i}^{n-1})\big(S_{y,j}^{n-1} + \tfrac{1}{2}(S_{y,j}^n - S_{y,j}^{n-1})\big), \tag{32}$$

$$\Delta S_{ij}^y = (S_{y,j}^n - S_{y,j}^{n-1})\big(S_{x,i}^{n-1} + \tfrac{1}{2}(S_{x,i}^n - S_{x,i}^{n-1})\big), \tag{33}$$

which satisfy $\Delta S^x + \Delta S^y = \Delta S$ identically. The edge currents are then the running sums

$$J_x\big|_{i+1/2,\,j} = -\frac{\text{value}}{\Delta t\,\Delta x}\sum_{k\le i}\Delta S_{kj}^x, \qquad J_y\big|_{i,\,j+1/2} = -\frac{\text{value}}{\Delta t\,\Delta x}\sum_{l\le j}\Delta S_{il}^y, \tag{34}$$

with value $= m$ (matter current) or q (charge current). Telescoping the sums reproduces Eq. (31) cell-by-cell. The trajectory endpoints use $\mathbf{x}^{n-1} = \mathbf{x}^n - \mathbf{v}^{n-1/2}\Delta t$, exact under the leapfrog drift. CIC (2×2 footprint) and bump variants are provided; if the Esirkepov support is exceeded (motion > 1 cell), the code falls back to a direct midpoint CIC current and flags the continuity violation.

10.3 Spin-dipole and smoothing

A spinning particle adds a divergence-free magnetization current. For a z-dipole moment μ deposited with shape W,

$$J_x = +\mu\,\partial_y W, \qquad J_y = -\mu\,\partial_x W \quad \Rightarrow \quad \nabla\cdot\mathbf{J} = 0, \tag{35}$$

so it perturbs $\mathbf{A}$ without adding to ρ (magnetic moment $\mu = g_s q/2m \cdot S$ for EM, gravitomagnetic moment $\sigma = 2S$). Finally, an optional binomial filter — the 3×3 stencil $\frac{1}{16}[1\,2\,1;\,2\,4\,2;\,1\,2\,1]$, applied N times to ρ (and to $\mathbf{J}$) — damps the residual grid-scale content of the deposit, the canonical PIC noise-control step. It is exposed as `rhoSmooth` / `jSmooth` pass counts.

11 Force Gather and the Lewis–Birdsall Scheme

Because the state is potentials, the force (9)–(10) is assembled at each particle from quantities gathered on the fly — no $\mathbf{g}, \mathbf{B}_g, \mathbf{E}, \mathbf{B}$ grid arrays are ever formed:

- $\nabla\Phi_g$ and $\nabla\phi$: gathered gradients of the corner scalars;
- $\partial_t\mathbf{A}_g,\ \partial_t\mathbf{A}$: centered time differences of the perturbation edge potentials, $(\Psi^{n+1} - \Psi^{n-1})/2\Delta t$;
- $B_{g,z}, B_z$: the Yee curl (§8) evaluated at the four cell centers around the particle and bilinearly interpolated;

each combining the analytic (or sampled) background with the grid perturbation.

11.1 Legacy vs. Lewis–Birdsall interpolation

Two gather styles are selectable. The *legacy* style interpolates field values with the shape W and finite-differences them. The *Lewis–Birdsall* (LB) style instead gathers the gradient *directly* with the analytic derivative of the same shape,

$$\partial_x\Psi(\mathbf{x}_p) = \frac{1}{\Delta x}\sum_{i,j}\frac{\mathrm{d}w}{\mathrm{d}u}(x_p - x_i)\,w(y_p - y_j)\,\Psi_{ij}, \tag{36}$$

and analogously for ∂_y and for the edge-field gathers entering $\mathbf{v} \times \mathbf{B}$. Pairing the derivative-shape gather (36) with the value-shape deposit is the adjoint (Hockney–Eastwood) condition at the gradient level: it makes the inter-particle force exactly reciprocal and cancels the grid self-force, so LB is the production choice. The 1PN gravitational vector $\mathbf{g} = -\nabla\Phi_g - \partial_t \mathbf{A}_g$ used in the EIH terms is built from the same gathered pieces.

11.2 The assembled 2D force

Collecting Part I, the per-particle force the pusher applies is (relativistic tier; $\mathbf{g}$ from the gathers above, $B_{g,z}, B_z$ from the Yee curl)

$$\boxed{\begin{aligned} F_x &= m\Big[(1 + \tfrac{v^2}{c^2} + \tfrac{4\Phi_g}{c^2})\, g_x - \tfrac{4(\mathbf{v}\cdot\mathbf{g})}{c^2} v_x + 4v_y B_{g,z} - \partial_t A_{g,x}\Big] + q\big[E_x + v_y B_z\big] + s\,\partial_x B_{g,z} + \mu\,\partial_x B_z, \\ F_y &= m\Big[(1 + \tfrac{v^2}{c^2} + \tfrac{4\Phi_g}{c^2})\, g_y - \tfrac{4(\mathbf{v}\cdot\mathbf{g})}{c^2} v_y - 4v_x B_{g,z} - \partial_t A_{g,y}\Big] + q\big[E_y - v_x B_z\big] + s\,\partial_y B_{g,z} + \mu\,\partial_y B_z, \end{aligned}}$$

with $E_\alpha = -\partial_\alpha\phi - \partial_t A_\alpha$, $\mu = g_s q/2m\cdot s$. Lower tiers drop the $\frac{v^2}{c^2}$, $\frac{4\Phi_g}{c^2}$, $(\mathbf{v}\cdot\mathbf{g})$ terms (EIH→GEM) and the $v \times B_g$ term (GEM→Newtonian); the EM and spin pieces are unaffected. Each gravitomagnetic cross product is in practice evaluated from the potential-form identity (§1.4) so $\mathbf{B}_g$ is never gridded.

12 The Relativistic Boris Pusher

12.1 Kick–drift with a velocity corrector

Particles advance by a relativistic Boris leapfrog (kick–drift):

$$\mathbf{p}^{n+1/2} = \mathbf{p}^{n-1/2} + \mathbf{F}^n \Delta t, \quad \mathbf{v}^{n+1/2} = \frac{\mathbf{p}^{n+1/2}}{\sqrt{m^2 + |\mathbf{p}^{n+1/2}|^2/c^2}}, \quad \mathbf{x}^{n+1} = \mathbf{x}^n + \mathbf{v}^{n+1/2}\Delta t. \tag{37}$$

The force depends on velocity (the v^2/c^2 EIH terms and the $\mathbf{v} \times \mathbf{B}$ terms), which is not known at t^n. GRlite uses the lagged half-step velocity $\mathbf{v}^{n-1/2}$ as a predictor, then, whenever the force is velocity-dependent (relativistic tier, or $B_{g,z} \neq 0$, or a charged particle in B_z), runs one corrector: predict $\mathbf{p}$, form the midpoint $\mathbf{v}_{\text{mid}} = \frac{1}{2}(\mathbf{v}^{n-1/2} + \mathbf{v}_{\text{pred}})$, and re-evaluate the force there. This restores second-order time accuracy and the time-symmetry of the push.

12.2 Self-field subtraction

For particles that opt in (§15), the gathered force includes the particle's own deposited field — a residual grid self-force. GRlite cancels it by maintaining a per-particle "self" field set (the same deposit and leapfrog, sourced by that particle alone) and forming

$$\mathbf{F}_{\text{total}} = \mathbf{F}_{\text{collective}} - (1 + \varepsilon)\,\mathbf{F}_{\text{self}}, \tag{38}$$

re-gathered through the identical chain; $\varepsilon = 0$ gives exact bit-level cancellation of the self-interaction, leaving only the inter-particle force.

12.3 Spin, proper time, and driven sources

After the push, the scalar spin phase advances by the 2D precession rate (Eq. (20), $\Omega_z = B_{g,z} + \frac{g_s q}{2m} B_z + \Omega_{\mathrm{Thomas},z}$ with $\Omega_{\mathrm{Thomas},z} = -\frac{\gamma^2}{\gamma+1}(v_x a_y - v_y a_x)/c^2$ and $\mathbf{a} = (\mathbf{v}^{n+1/2} - \mathbf{v}^{n-1/2})/\Delta t$), and the proper-time clock by

$$\Delta\tau = \Delta t \sqrt{1 + \frac{2\Phi_g}{c^2} - \left(1 - \frac{2\Phi_g}{c^2}\right)\frac{v^2}{c^2} - \frac{8\,\mathbf{A}_g\cdot\mathbf{v}}{c^2}} \tag{39}$$

(Eq. (21)). Two source modes override the plain push: a *pinned* particle (forces disabled) does not integrate the force, acting as a fixed emitter; a *driven* particle carries a base velocity $\mathbf{v}_{\mathrm{base}}$ (itself evolved under the force) plus a sinusoidal modulation $\mathbf{v} = \mathbf{v}_{\mathrm{base}} + \mathbf{v}_{\mathrm{drive}} \cos(\omega t + \varphi)$, written into the momentum so the Esirkepov current stays charge-conserving. Storing the base separately is what prevents the oscillation from accumulating into the momentum — the model for an antenna/dipole emitter.

13 Initial Conditions

Starting the leapfrog from zero fields with sources already present injects a spurious "turn-on" transient: the field has to radiate out to its static configuration, and the standing modes of a finite box are slow to leave. GRlite seeds a near-static initial field instead, by two ingredients:

Liénard–Wiechert seed. The potentials are initialized to the direct-sum (here static) Liénard–Wiechert superposition of the particle sources, with a boundary-mean taper that matches the interior solution smoothly to the box edge (matched to zero across the damping ring by default). This places the field within a small residual of the discrete static solution at $t = 0$.

Frozen-particle friction settle. The L–W seed is then relaxed to the exact discrete fixed point by running N_{settle} steps with the particles *frozen* (sources still deposit from the frozen state; the field still evolves) and the derivative-friction absorber (§14) on. The friction drains the residual standing modes to round-off in a few hundred steps, leaving the field at the discrete Poisson solution of the deposited ρ.

These compose into the scenario-level initialization choices `lw-settled` (seed + settle, the default for interacting runs), `lw` (seed only), `none` (auto: settle when a perturbation is active, else leave at zero), and `zero` (always start from a quiet field — used by wave-emission/antenna demos that must not begin with a static well). A direct Jacobi/Poisson relaxation of Φ_g is also available as an alternative seed.

14 Boundary Conditions and Gauge Control

14.1 Outer boundary

The non-periodic solver applies, after each leapfrog sweep, either a **Dirichlet** ring ($\Psi = 0$ on the outer cells: waves reflect with a sign flip; the discrete static state is the zero-Dirichlet Poisson solution) or a **Neumann** ring ($\Psi_{\mathrm{edge}} = \Psi_{\mathrm{inward}}$: reflection without sign flip, putting standing-mode antinodes at the wall where the absorber is strongest, and better approximating the 2D free-space $\ln r$ Green's function). A fully **periodic** variant (wrap-around indexing) is also available.

14.2 The absorbing layer

Outgoing waves are removed by a derivative-friction layer implementing the damped wave equation $\partial_t^2\Psi + 2\gamma\,\partial_t\Psi = c^2\nabla^2\Psi + \ldots$. The discretization is *centered-implicit*: the standard leapfrog `next` is corrected, before the buffer rotation, by

$$\Psi_k^{n+1} \leftarrow \frac{\Psi_k^{n+1} + \beta_k\,\Psi_k^{n-1}}{1+\beta_k}, \qquad \beta_k = \gamma_k\,\Delta t. \tag{40}$$

At the 2D Nyquist mode the recurrence roots are -1 (marginal) and $-(1-\beta)/(1+\beta)$ (stable), so (40) is stable for *any* β — unlike a naive bleed on `prev`, which destabilizes the Nyquist mode at $\mathrm{cfl} = 1/\sqrt{2}$. The coefficient β_k is a cubic ramp from a small interior *floor* (which also drains the lowest box mode) up to a *wall* value over a taper *depth* of cells at the boundary.

14.3 Gauge maintenance

Two optional fixes keep the Lorenz gauge from drifting. The **zero-mean** fix addresses the unconstrained DC mode of the scalar potentials under Neumann BC (the wave equation accelerates it whenever the source mean is nonzero): subtract the spatial mean from both `curr` and `prev` each step (equal subtraction preserves the encoded time derivative). The **parabolic Lorenz cleaning** subtracts a multiple of the gauge residual, $\phi \mathrel{-}= R\,c^2\Delta t\,(\nabla\cdot\mathbf{A})$ (and likewise Φ_g with $\nabla\cdot\mathbf{A}_g$), bounding the drift introduced by sources whose discrete divergence is not exactly consistent (e.g. the spin-dipole curl current).

15 Background Field Installation

The fixed background of §5 is installed from a list of up to sixteen compact bodies, each with parameters $(x, y, GM, Q, J_z, g_s, \varepsilon)$. The list is the source of truth; the field is derived from it in one of two modes.

Sampled mode. Each body's analytic generators (§5.2) — the softened gravitational scalar, the gravitomagnetic dipole, the Coulomb scalar, and (for a spinning charge) the EM magnetic dipole — are sampled *additively* onto the six background grid arrays, on each potential's own sublattice. Multiple bodies superpose by accumulation, which is exact in the linear theory. The force gather then reads these grids exactly as it reads the perturbation, and the Shapiro c_{local} (Eq. (12)) is built from the *total* Φ_g (sampled background + perturbation), so a deposited mass lenses light on top of the fixed body.

Analytic mode. The closed-form fields are instead evaluated directly at each particle position and summed over the bodies, with no grid-sampling error. This eliminates the $\mathcal{O}((\Delta x/r)^2)$ tangential-force error of the sampled path that would otherwise induce a slow grid precession — so analytic mode is preferred for clean test-particle orbits, while sampled mode is used when the background must enter the gridded field evolution (e.g. dynamic lensing). Either way, a single body reduces exactly to the corresponding classical-metric analogue of §5.2.

16 Visualization Mathematics

The visualization is part of the physics: a pedagogical and debugging tool only works if the rendered quantity is exactly the simulated one. All field views are computed CPU-side each frame from the potential arrays and the chosen *source* (perturbation only, background only, or their sum).

Field colormap. A selectable scalar is mapped through a diverging colormap auto-normalized to the current frame's peak magnitude (so the scale tracks the dynamics). The available reductions are the stored scalars themselves (Φ_g, ϕ); a *curl* field $B_z = \partial_x A_y - \partial_y A_x$ (and $B_{g,z}$) formed from the two edge arrays; and a *gradient magnitude* $|\nabla\Phi_g|$ for the field strength.

Vector overlay (quiver). On a coarse sub-grid the overlay draws either edge-potential components ($\mathbf{A}_g, \mathbf{A}$) or a gradient field ($\mathbf{g} = -\nabla\Phi_g$, $\mathbf{E} = -\nabla\phi$), arrows normalized so the strongest spans ≈ 0.9 of the sample spacing, with opacity scaling by relative magnitude so weak regions recede.

Per-particle overlays. Each maps a simulated quantity to a glyph:

- **velocity arrow** of pixel length $L_v\,|\mathbf{v}|$ with $\mathbf{v} = \mathbf{p}/(\gamma m)$ (a fixed pixels-per-c scale), brightened above $|\mathbf{v}| > 0.5c$;
- **force arrows** for the gravitational, EM, spin, and total components (the per-particle breakdown stashed by the pusher), drawn in distinct colors and sharing *one* auto-scale (the largest component across all particles spans a fixed reference length) so magnitudes are directly comparable;
- **spin indicator**: a hand through the particle at angle ϕ_{spin}, so Lense–Thirring/Thomas precession is literally visible;
- **trajectory ticks**: along the recorded track, circles at equal *coordinate*-time intervals Δ and triangles at equal *proper*-time intervals Δ (auto-chosen to ~ 12 ticks across the trail). Where the proper-time triangles spread apart relative to the coordinate circles, time dilation is being shown directly;
- **proper-time clock**: a two-hand dial, one hand sweeping coordinate time t and one proper time τ at a common period; the opening wedge between them is the accumulated dilation $t - \tau$;
- **background body**: a hollow ring sized to the softening length ε at the body's position.

Every overlay is independently toggleable so a single effect can be isolated.

End of Part II. Part III documents the C library that implements the scheme above and maps each public function to the pipeline step (§7) it realizes.

Part III
The C Library

The simulation core is a freestanding C11 library (`core/`, compiled both natively for the test suite and to WebAssembly for the browser). It has no dependencies beyond `libm`. This Part documents the data structures that hold the state of Part II, the source files that implement each piece of the scheme, and the complete public API with each function tied back to the algorithm step it realizes. The single public header is `core/include/grlite.h`; the internal state struct lives in `core/src/sim_internal.h`.

17 Architecture and Data Structures

17.1 The simulation object

A run is a single opaque handle `gr_sim_t*`, created with the grid and the effective constants of Part I:

```
gr_sim_t* gr_sim_create(int width, int height, float dx, float c_eff, float cfl);
```

Internally (`struct gr_sim`) it holds the entire state of §7:

Member	Role (Part II reference)
`c_eff, dt, cfl, dx`	wave speed / step / Courant ratio (Eq. (28))
`G_eff, k_e`	effective couplings G, k_e (Part I)
`fields[6]`	the six potentials, each a `gr_field_state_t` (§8)
`rho_matter, J_mx, J_my`	gravity sources $\rho_{\mathrm{matter}}, \mathbf{J}_{\mathrm{matter}}$ (§10)
`rho_q, J_qx, J_qy`	EM sources $\rho_q, \mathbf{J}_q$
`particles, n_particles`	the macroparticle array (§12)
`damping_d, friction_d`	absorbing-layer arrays (§14)
`phi_g_bg, ..., Ay_bg`	sampled background fields (§15)
`bg_bodies[16], bg_nbody`	analytic compact-body list (§15)
`self_field_sets`	per-particle self-field sets (Eq. (38))
`force_tier, shape_function, force_interp`	scheme selectors (§§9,11)
`*_enabled flags`	the physics switches (force terms, Shapiro, gauge)

17.2 Field, particle, and body records

Each potential is a three-time-level record on its Yee sublattice; the leapfrog (27) reads `prev`/`curr`, writes `next`, and the step rotates the three pointers:

```
typedef struct { float *prev, *curr, *next;     /* Phi^{n-1}, Phi^n, Phi^{n+1} */
                 float *source; float source_coeff; } gr_field_state_t;
```

`source` binds to one of the six source arrays and `source_coeff` is the s_Ψ of Eq. (27). A macroparticle carries the dynamical state of §12 plus per-step diagnostics:

```
typedef struct { float x, y, px, py;            /* position, relativistic momentum */
                 float mass, charge, proper_time;
                 float spin, phi_spin, g_factor; /* scalar spin (Sec. 4.3) */
                 float fgrav_x..ftot_y;          /* force breakdown (viz, Sec. 16) */
                 float drive_vx..drive_base_vy;  /* driven/pinned source (Sec. 12) */
               } gr_particle_t;
```

and a background body is the seven contiguous floats of the no-hair source (§5.2):

```
typedef struct { float x, y, GM, Q, Jz, g_em, eps; } gr_bg_body_t;
```

17.3 Memory layout and access

Every grid array is a flat `float[W*H]` in row-major order (index $k = jW + i$), matching the leapfrog indexing of Eq. (27). All grids share one geometry; only the node interpretation (corner / edge) differs by sublattice. Because the arrays and the particle records are plain `float`s in a single linear heap, the WebAssembly build exposes them to the front end as zero-copy `Float32Array` views: `gr_sim_field_ptr`, `gr_sim_array_ptr`, `gr_sim_background_ptr`, and `gr_sim_get_particle` hand back interior pointers that the visualization (§16) reads directly each frame.

17.4 Lifecycle

The driver loop is just `gr_sim_create` → configure → add particles/background → initialize fields (§13) → `gr_sim_step`/`gr_sim_step_n` → `gr_sim_destroy`. The convenience entry `gr_sim_use_pedagogical_defaults` installs the production bundle (bump kernel, Lewis–Birdsall, deposition on, absorbing layer) in one call.

18 Module-by-Module Correspondence

The implementation is six source files, each owning one band of Part II (Table 4).

File	Implements (Part II)	Key symbols
`sim.c`	lifecycle; the per-step driver (§7); init (§13); friction taper, zero-mean (§14); config	`gr_sim_step`, `_create`, `_init_potentials_settled`, `_set_volume_friction_taper`
`field.c`	leapfrog + variants (§8); parabolic gauge clean (§14)	`gr_field_leapfrog_step_all`, `_step_self`, `gr_field_parabolic_gauge_clean`
`deposit.c`	shapes (§9); Esirkepov (§10); LB gathers (§11); spin-dipole curl	`gr_{cic,tsc,bump}_deposit_corner`, `gr_{,bump_}esirkepov_deposit_jxy`, `gr_bump_lb_grad_corner`
`particle.c`	force gather (§11); Boris push, corrector, self-field, spin, proper time, drive (§12)	`gr_particle_push_all`, `gr_phi_g_total_at`
`background.c`	compact-body generators + analytic evaluators (§15); Shapiro c_{local} rebuild (§2)	`gr_sim_{set,add}_background_body`, `gr_bg_eval_*`, `gr_em_shapiro_recompute_c_local2`
`gauge.c`	Lorenz-gauge residual diagnostics (§14)	`gr_sim_gauge_residual_{grav,em}`

Table 4: Source-file to Part II mapping.

Two correspondences are worth spelling out because they thread several files.

The step driver (`sim.c` ⇒ §7). `gr_sim_step` executes the pipeline literally in order: it clears the source arrays and, per particle, calls the selected `gr_*_deposit_corner` for ρ^n and `gr_*_esirkepov_deposit_jxy` for $\mathbf{J}^{n-1/2}$ (with `gr_*_curl_dipole_deposit_jxy` for spin), optionally binomial-smooths them, folds in EM stress-energy, refreshes c_{local} (`background.c`), calls `gr_field_leapfrog_step_all` (`field.c`), applies the centered-implicit friction (Eq. (40)) to each `next` before the pointer rotation, runs the gauge fixes, and finally calls `gr_particle_push_all` (`particle.c`) unless the particles are frozen.

The push (`particle.c` ⇒ §§11–12). `gr_particle_push_all` gathers each force component through the static evaluators (`grav_grad_at/_lb`, `B_g_z_at_total`, `phi_em_grad_at_total`, `dt_A_*_at_total`), assembles the tiered force, runs the velocity corrector, performs the self-field subtraction by temporarily swapping `sim->fields` for the particle's `self_field_set` and replaying the same gather, then updates momentum, spin phase, proper time, and position. The force-component fields it writes back (`fgrav/fem/fspin/ftot`) are exactly what the force-arrow overlay (§16) renders.

19 API Reference

The public API (`grlite.h`) is listed below by function group, each entry tied to the Part II step it serves. Arguments are abbreviated; `s` denotes the `gr_sim_t*` handle. Getters that merely return the value set by their paired setter are folded into one row ("set/get").

19.1 Lifecycle and time

Function	Role	Part II
`gr_sim_create(w,h,dx,c_eff,cfl)`	allocate a run	§17
`gr_sim_destroy(s)`	free it	§17
`gr_sim_use_pedagogical_defaults(s)`	install production bundle	§17
`gr_sim_step(s)` / `gr_sim_step_n(s,n)`	advance the pipeline	§7
`gr_sim_step_count(s)`, `gr_sim_time(s)`	step index, $t = n\,\Delta t$	§7
`gr_sim_reset_time(s)`	zero the step counter	§7
`gr_sim_dt/dx/width/height(s)`	grid/time accessors	§8

19.2 Couplings and scheme selectors

Function	Role	Part II
`gr_sim_set/get_G_eff(s,` $\cdot$`)`	gravity coupling G	Part I
`gr_sim_set/get_k_e(s,` $\cdot$`)`	EM coupling k_e	Part I
`gr_sim_set/get_force_tier(s,` $\cdot$`)`	Newtonian / GEM / EIH	§11
`gr_sim_set/get_shape_function(s,` $\cdot$`)`	CIC / TSC / bump	§9
`gr_sim_set/get_kernel_radius(s,` R`)`	bump radius R	§9
`gr_sim_set/get_force_interp(s,` $\cdot$`)`	legacy / Lewis–Birdsall	§11

19.3 Physics switches

Each toggles one term of the assembled force, source, or gauge machinery; all are `set/get` pairs taking `(s, enabled)`.

Function (`gr_sim_set_···_enabled`)	**Gates**	**Part II**
`gravitomagnetic_force`	$4m\,\mathbf{v}\times\mathbf{B}_g$	§11
`gravitomagnetic_inductive`	$-m\,\partial_t\mathbf{A}_g$	§11
`em_lorentz_force`	$q(\mathbf{E}+\mathbf{v}\times\mathbf{B})$	§11
`em_inductive`	$-q\,\partial_t\mathbf{A}$	§11
`em_electrostatic`	$-q\nabla\phi$	§11
`em_magnetic`	$q\,\mathbf{v}\times\mathbf{B}$	§11
`em_stress_energy`	$\rho_{\mathrm{EM}},\mathbf{J}_{\mathrm{EM}}$ source	§7
`em_shapiro`	c_{local} EM solve	§2
`em_shapiro_dynamic`	per-step c_{local} rebuild	§15
`parabolic_gauge_cleaning`	Lorenz cleaning	§14
`field_evolution`	run the leapfrog at all	§8
`particle_source_deposition`	deposit particle sources	§10
`esirkepov`	charge-conserving $\mathbf{J}$	§10
`periodic_bc`	wrap-around boundary	§14
`particles_frozen`	freeze the push (settle)	§13

19.4 Sources and deposition

Function	**Role**	**Part II**
`gr_sim_clear_sources(s)`	zero the six source arrays	§7
`gr_sim_clear_fields(s)`	zero all potentials	§13
`gr_sim_deposit_point_mass/charge(s,x,y,·)`	direct ρ deposit (shape)	§9
`gr_sim_deposit_point_particle(s,...)`	deposit ρ and $\mathbf{J}$	§10
`gr_sim_rho_matter_ptr/rho_q_ptr(s)`	source-array views	§10
`gr_sim_J_mx/my/qx/qy_ptr(s)`	current-array views	§10
`gr_sim_set/get_rho_smooth_passes(s,n)`	ρ binomial smoothing	§10
`gr_sim_set/get_j_smooth_passes(s,n)`	$\mathbf{J}$ binomial smoothing	§10
`gr_sim_esirkepov_violations(s)`	continuity-fallback count	§10

19.5 Field / array access and gauge diagnostics

Function	**Role**	**Part II**
`gr_sim_field_ptr(s,which)`	a potential array (`prev/curr`)	§8
`gr_sim_array_ptr(s,which)`	any field/source array	§17
`gr_sim_background_ptr(s,which)`	a sampled background array	§15
`gr_array_lattice(which)`	sublattice of an array	§8
`gr_lattice_offset(lat,·,·)`	node half-cell offset	§8
`gr_sim_gauge_residual_grav/em(s)`	$\lVert\partial_\mu\bar{h}^{\mu\nu}\rVert$ monitor	§14

19.6 Particles

Function	Role	Part II
`gr_sim_add_particle(s,x,y,m,q,vx,vy)`	append a macroparticle	§12
`gr_sim_particle_count(s)`	number of particles	§12
`gr_sim_get_particle(s,idx)`	read a particle record	§16
`gr_sim_clear_particles(s)`	remove all	§12
`gr_sim_particle_energy(s,idx)`	γmc^2 diagnostic	§12
`gr_sim_set_particle_spin(s,idx,s,g)`	scalar spin + g_s	§4.3
`gr_sim_set_particle_phi_spin(s,idx,`ϕ`)`	spin phase	§4.3
`gr_sim_set_particle_drive(s,idx,...)`	driven/pinned source	§12
`gr_sim_particle_enable/disable_self_field(s,idx)`	toggle self-field set	Eq. (38)
`gr_sim_particle_has_self_field(s,idx)`	query	Eq. (38)
`gr_sim_particle_set/get_self_field_ epsilon(s,idx,·)`	subtraction ε	Eq. (38)

19.7 Initialization

Function	Role	Part II
`gr_sim_relax_phi_g_poisson(s,n)`	Jacobi Poisson seed	§13
`gr_sim_init_potentials_lienard_wiechert(s)`	L–W seed	§13
`gr_sim_init_potentials_lienard_wiechert_with_ taper(s,in,out)`	L–W seed + boundary taper	§13
`gr_sim_init_potentials_settled(s,n_settle)`	L–W + frozen friction settle	§13

19.8 Boundary, absorber, and damping

Function	Role	Part II
`gr_sim_set/get_outer_bc_neumann(s,·)`	Dirichlet vs Neumann ring	§14
`gr_sim_set_volume_friction_ taper(s,floor,wall,depth)`	derivative-friction absorber	§14
`gr_sim_volume_friction_enabled(s)`	query	§14
`gr_sim_set/get_zero_mean_scalar_potentials(s,·)`	DC-mode gauge fix	§14
`gr_sim_set_damping(s,n) / _damping_layers(s)`	PML width	§14
`gr_sim_set/get_damping_config(s,cfg)`	PML profile params	§14
`gr_sim_damping_max_stable_sigma_dt(s,form)`	stability bound	§14

19.9 Background fields

The unified compact body and the multi-body calls implement §15; the uniform/linear setters are single-body isolation fixtures.

Function	Role	Part II
`gr_sim_clear_background(s)`	drop all background	§15
`gr_sim_set_background_body(s,x,y,GM,Q,Jz,g,eps)`	one (M, Q, J) body	§5.2
`gr_sim_add_background_body(s,...)` $\rightarrow$ idx	append a body (superpose)	§15
`gr_sim_background_count(s)`	number of bodies	§15
`gr_sim_get_background_body_ptr(s,i)`	body params view (7 floats)	§15
`gr_sim_set_background_body_at(s,i,...)`	edit body i (re-derive)	§15
`gr_sim_set_background_point_mass/_charge(s,...)`	Schwarzschild / RN special case	§5.2
`gr_sim_set_background_spinning_point_mass(s,...)`	Kerr special case	§5.2
`gr_sim_set_background_uniform_gravitomagnetic/_ magnetic/_electric(s,...)`	uniform test fields	§15
`gr_sim_set_background_linear_magnetic(s,...)`	$B_z(x)$ gradient fixture	§15
`gr_sim_set/get_bg_mode(s,·)`	sampled vs analytic	§15
`gr_sim_get_bg_kind(s)`	installed background kind	§15

19.10 Scenario registry

Function	Role	Part II
`gr_sim_load_scenario(s,name,params,n)`	build a registered C scenario	§7
`gr_scenario_register(def)`	register one	—
`gr_scenarios_init()`	populate the registry	—

20 The Scenario Layer and Web Front End

A run's initial conditions can be assembled two ways, both built on the API of §19. The native path is the C *scenario registry* (`gr_sim_load_scenario`): each test in the suite is a registered scenario plus an analytic assertion, which is how the physics of Parts I–II is regression-checked.

The browser path is declarative: a JSON object — `{program, format, version, grid, global, background, particles, view}` — is the single source of truth, and the front end's `applyScenario` routine maps it onto exactly the calls documented above:

JSON field	API calls
`grid`	`gr_sim_create` (W,H,dx,cEff,cfl)
`global.gEff/kE/...`	couplings + scheme selectors (§19)
`global.switches.*`	the physics-switch setters (one-to-one)
`global.absorber/init`	friction taper + the `init_potentials_*` calls
`background[]`	`gr_sim_clear_background` + `gr_sim_add_background_body` per body
`particles[]`	`gr_sim_add_particle` + `set_particle_spin/drive`, self-field
`view`	front-end visualization only (§16)

The JSON is thus a thin, validated, shareable wrapper over the same C entry points; a scenario fully and reproducibly determines a run. A debug bridge additionally exposes the API over a local socket, returning *computed* diagnostics (field statistics, conservation residuals, per-particle state) so the simulation can be driven and checked programmatically.

21 Conclusion

The three parts form one chain of custody for the model. Part I states the physics — linearized GR in GEM form coupled to electromagnetism, with relativistic, spin, and proper-time dynamics — deriving each result in 3D and specializing it to the 2+1 dimensions the code runs in. Part II discretizes those 2D equations into an explicit Yee-lattice PIC scheme: leapfrog field advance, shape-function deposition, Esirkepov charge conservation, Lewis–Birdsall force gather, a relativistic Boris push, and the initial-field, boundary, background, and visualization machinery. Part III ties that scheme to the C library, member by member and function by function. The cross-references run the full length: an equation label in Part I, an algorithm step in Part II, and a function in Part III name the same object at three levels of description. That is the intent — for a simulator that borders on research work, the work is shown.

End of report.

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